

Task 1 Dockerize DevOps Control Plane

Objective

Containerize the DevOps Control Plane Streamlit application with persistent JSON logging, health checks, and a named Docker volume for logs. The app runs on Streamlit's default port 8501 and is designed to be started through Docker Compose.

What was built

The application includes a Streamlit dashboard, system metrics collection, structured JSON logging, health checks, and container packaging. The project also uses a multi-stage Dockerfile, a `.dockerignore` file, and a Docker Compose setup with the `logsvolume` named volume mounted at `/applogs`.

Project structure

- `app.py` or `uidashboard.py` for the Streamlit UI.
- `coremetrics.py` for CPU, RAM, and disk metrics.
- `corehealth.py` for readiness and liveness checks.
- `coreconfig.py` for environment-driven settings.
- `infralogger.py` for JSON structured logging.
- `requirements.txt` for Python dependencies.
- `Dockerfile` for the container image.
- `.dockerignore` to reduce build context.
- `docker-compose.yml` for local startup with a persistent log volume.
- `README.md` for short usage instructions.

Commands used

```
docker compose up --build

docker compose up -d --build

docker compose down

docker compose logs -f

docker compose build --no-cache

docker ps

curl http://localhost:8501/_stcore/health
```

These commands were used to build, run, stop, inspect, and verify the containerized app.

Configuration details

- Port: 8501, kept as Streamlit's default web port.
- Volume: `logsvolume`, used for persistent application logs.
- Log location inside container: `/applogs`.
- Environment variables: `LOGLEVEL` and `REFRESHRATE`.
- Health check: Streamlit health endpoint at `http://localhost:8501/_stcore/health`.

Troubleshooting notes

A common issue during Task 1 was missing Python package structure files such as `__init__.py`, which can cause `ModuleNotFoundError` when importing modules from `core`, `infra`, or `ui`. Another issue was ensuring the Docker volume and log directory existed correctly so the logger could write JSON logs without permission problems.

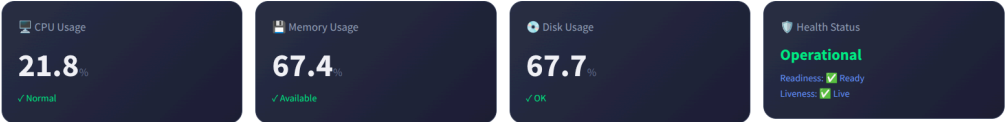
Verification

The container was verified by checking that the Streamlit app was reachable in the browser and that the health endpoint responded successfully. Logs were also confirmed to be written into the persistent log volume.

Screenshot placeholder

Deploy

System metrics & structured logging - Auto-refresh every 5s



System Details

Current Load
CPU: 21.8% - RAM: 67.4% - Disk: 67.7%

Last Update
2025-06-09 09:55:50

Force Refresh

Structured Logs (Last 20 entries)

No logs found. Waiting for application logs...

Made with Streamlit

Auto-refresh every 5s - 09:55:50

File Edit Selection View Go Run Terminal Help

harena-devops-labs

PROBLEMS (145) OUTPUT DEBUG CONSOLE TERMINAL PORTS SONARQUBE

```
> docker ps
CONTAINER ID   IMAGE                                     COMMAND                  CREATED        STATUS        PORTS
23502e1f20d6   to-malagasy-translate-api-api          "unicorn -c gunicor..." 23 hours ago   Restarting (1) 14 seconds ago
193af0b37b7c   postgres:15-alpine                    "docker-entrypoint.s..." 23 hours ago   Up 2 hours   5432/tcp
0dee77c8c5f2   to-malagasy-translate-api-postgres-1   "docker-entrypoint.s..." 23 hours ago   Up 2 hours   6379/tcp
a3136b420681   ghcr.io/open-webui/open-webui:main     "bash start.sh"          6 days ago    Up 2 hours (healthy)   0.0.0.0:3000->8080/tcp, [::]:3000->8080/tcp
4b95fa1f380f   ollama/ollama:latest                  "/bin/ollama serve"      5 weeks ago   Up 2 hours   0.0.0.0:11434->11434/tcp, [::]:11434->11434/tcp
> docker ps -a
CONTAINER ID   IMAGE                                     COMMAND                  CREATED        STATUS        PORTS
23502e1f20d6   to-malagasy-translate-api-api          "unicorn -c gunicor..." 23 hours ago   Restarting (1) 33 seconds ago
193af0b37b7c   postgres:15-alpine                    "docker-entrypoint.s..." 23 hours ago   Up 2 hours   5432/tcp
0dee77c8c5f2   to-malagasy-translate-api-postgres-1   "docker-entrypoint.s..." 23 hours ago   Up 2 hours   6379/tcp
a3136b420681   ghcr.io/open-webui/open-webui:main     "bash start.sh"          6 days ago    Up 2 hours (healthy)   0.0.0.0:3000->8080/tcp, [::]:3000->8080/tcp
4b95fa1f380f   ollama/ollama:latest                  "/bin/ollama serve"      5 weeks ago   Up 2 hours   0.0.0.0:11434->11434/tcp, [::]:11434->11434/tcp
71862ba808f1   postgres:15-alpine                    "docker-entrypoint.s..." 7 weeks ago    Exited (0) 7 weeks ago
sync-note-db

> docker build -t nyharena/harena-devops-labs:latest ../task1-dockerize
[+] Building 206.6s (12/12) FINISHED
docker:default
=> [internal] load build definition from Dockerfile
0.0s
=> transferring dockerfile: 460B
0.0s
=> [internal] load metadata for docker.io/library/python:3.11-slim
2.5s
=> [internal] load .dockerignore
0.0s
=> transferring context: 126B
```

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```
File Edit Selection View Go Run Terminal Help ← → harena-devops-labs
PROBLEMS 145 OUTPUT DEBUG CONSOLE TERMINAL PORTS SONARQUBE
⇒ [internal] load build definition from Dockerfile
   0.0s
⇒ => transferring dockerfile: 460B
   0.0s
⇒ [internal] load metadata for docker.io/library/python:3.11-slim
   2.5s
⇒ [internal] load .dockerignore
   0.0s
⇒ => transferring context: 126B
   0.0s
⇒ [internal] load build context
   0.0s
⇒ => transferring context: 18.08kB
   0.0s
⇒ [builder 1/4] FROM docker.io/library/python:3.11-slim@sha256:a3ab0b966bc4e91546a033e22093cb840908979487a9fc0e6e38295747e49ac0
   0.1s
⇒ => resolve docker.io/library/python:3.11-slim@sha256:a3ab0b966bc4e91546a033e22093cb840908979487a9fc0e6e38295747e49ac0
   0.0s
⇒ CACHED [builder 2/4] WORKDIR /app
   0.0s
⇒ [builder 3/4] COPY requirements.txt .
   0.0s
⇒ [builder 4/4] RUN pip install --no-cache-dir --user -r requirements.txt
  141.2s
⇒ [stage-1 3/5] COPY --from=builder /root/.local /root/.local
   3.4s
⇒ [stage-1 4/5] COPY . .
   0.1s
⇒ [stage-1 5/5] RUN mkdir -p /app/logs
   0.3s
⇒ exporting to image
  56.5s
⇒ => exporting layers
   48.8s
⇒ => exporting manifest sha256:7b21d5412700ed2c4e482405ec2823685688b25d96f2527df48e7b6cdee07228
   0.0s
⇒ => exporting config sha256:0ba59dd87fad23ca4b58243b90406d89827c4cf99b42a14e2ed61e9fafb24fa6
   0.0s
⇒ => exporting attestation manifest sha256:47ba208b51d2748f0eb90cfbc6b0e54c5d9eba27edf38a95ac834ff6ccc3adbc
   0.0s
⇒ => exporting manifest list sha256:1a8bc3661b17f22f3d73313790b55f4b86104b69bdbc97d980b87d57458e3ee
   0.0s
main* 119 26 Git Graph SonarQube
zaisan25 (now) Ln 15, Col 30 Spaces: 2 UTF-8 LF ( ) YAML Continue (NE)
```